From Completion Signals to Useful Prefetch

Separate remaining work, scheduling opportunity and physical data dependencies

Child observations
Notice, text, decode progress
Already delivered information

→

Phase / work model
Final report or further tool use
Conditional remaining work

→

Advisory output
Stage score + remaining-token bounds
Observed EOS: separate signal

Tool observations
Type, elapsed time, history
History of completed calls

→

Residual-time model
Given the tool is still running
Update as observations arrive

→

Advisory output
Residual CDF / timing bounds
No learned net-benefit decision

(a) Runtime turns hints into bounded actions

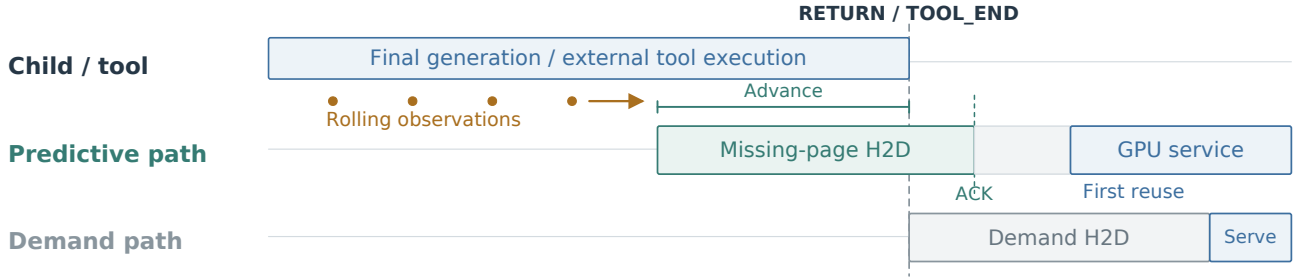
Causal readiness
JOIN / tool state

Reusable checkpoint
Valid missing Host pages

Capacity and cost
FULL / Mamba / H2D

Service opportunity
Admission + residency

(b) Overlap with the observed wait or final-generation interval



Schematic, not measured speedup. Partial overlap is useful; native data dependencies govern execution.